

# Roman Brunetti

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## EDUCATION

**B.S. Electrical Engineering** | North Carolina State University **Dec. 2027**

**Relevant Coursework:** Control Systems, Linear Systems, Electromechanical Energy Conversion, Engineering Dynamics, Microelectronics, Computer Systems Programming, Fundamentals of Logic Design, Calculus III

## PROJECTS

### Helping Hand Project: Robotic Prosthetic Hand

- Electrical Hardware Lead within a university-funded, multidisciplinary organization
- Designing power electronics to support servo-stall current transients while maintaining logic signal integrity
- Using OrCAD X PSpice simulation to target voltage averages and ripples
- Reading datasheets and selecting electrical and mechanical components to meet design tolerances

### FPGA Reciprocal Cipher

- Built FPGA circuit to take a 4-bit input, a 4-bit key, and output a 4-bit cipher
- Designed and implemented two-way analog-to-digital converter logic
- Implemented several layers of toggleable reciprocal ciphers using Verilog language and Apio tool
- Used lab equipment to analyze and debug circuit

### Precision Catapult

- Designed and fabricated a mechanical launch system with repeatable and calculable trajectories
- Designed adjustable launch geometry ( $0^{\circ}$ – $90^{\circ}$ ) to enable full coverage within the system's maximum range
- Devised a two-axis guide system to replicate radial guidance without specialized manufacturing

## WORK EXPERIENCE

**Orchard Inn Restaurant, Saluda NC** | Fine Dining Server & Line Cook June **2022-Present**

- Built strong working relationships over 3+ years with the same team

## SKILLS

### Programming

- C, Verilog, MATLAB, LC-3 assembly language, Git and GitHub

### Electrical

- Laplace techniques, impedance matching, transistor analysis, RLC circuits, three-phase, FPGA logic,
- PSpice (OrCAD X), basic PCB layout (Altium), breadboarding, through-hole soldering, multimeter, oscilloscope

### Mechanical

- Rotating and translating dynamic system modeling, internal force analysis, SolidWorks, 3D printing, hand and power tools

### Interests

Snowboarding | Mycology | Mountain biking | Tennis | Hiking | Camping | Piano | Cooking | Bookbinding | Reading | Woodworking